



# Gauge Historical Daily Synthetic Generation Model

Stochastic Water Level Timeseries with Tidal Classification

MKM Research Labs

Version 1.0 — 18-March-2026

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# 1 Executive Summary

This document presents the mathematical framework, implementation, and validation of the Gauge Historical Daily (GaugeHD) Synthetic Generation Model (MKM-GHD-001). The model generates realistic daily water level timeseries for flood gauges in the Thames catchment, producing 50-year synthetic histories that serve as the baseline for downstream hazard analysis, stress testing, and PRS pricing.

The central design objective is to produce daily water levels that are statistically consistent with the gauge’s flood thresholds (FloodAlert, FloodWarning, SevereFloodWarning), historical high records, and geographic tidal characteristics. The model classifies each gauge into one of three tidal regimes—*Non-tidal*, *Partially tidal*, or *Tidal*—based on the gauge’s longitude relative to Teddington Lock, and applies regime-specific parameters controlling seasonal amplitude, tidal oscillation, daily noise, and autocorrelation.

Five stochastic components are combined additively: (i) a base level calibrated to the gauge’s FloodAlert threshold, (ii) a seasonal cosine with winter peak, (iii) a spring-neap tidal cycle for tidally-influenced gauges, (iv) an AR(1) autocorrelated noise process, and (v) Poisson-distributed flood event injection. Historical high levels are placed at their recorded dates when available.

**Key risk:** The synthetic timeseries feeds the monthly means computation used by the stress model pipeline (stressm) and provides the historical baseline for GEV fitting. An incorrect tidal classification or mis-calibrated autocorrelation coefficient will distort the variance structure of the generated series, leading to biased flood frequency estimates downstream.

## 2 Model Purpose and Scope

### 2.1 Purpose

The GaugeHD Synthetic Generation Model serves three principal purposes within the Physical Risk platform:

1. **Historical baseline generation.** Provide a 50-year daily water level timeseries for each gauge, establishing the distributional baseline from which flood exceedance statistics are derived.
2. **Monthly means for stress testing.** The generated timeseries is summarised into monthly mean water levels, which are consumed by the stress model pipeline (**stressm**) as a baseline for scenario construction.
3. **Level statistics.** Compute percentiles, annual maxima, flood exceedance counts, and seasonal patterns that inform gauge-level hazard curve calibration and validation.

### 2.2 Scope

- **In scope:** Univariate daily water level generation; tidal classification; seasonal, tidal, and AR(1) noise components; Poisson flood injection; historical high date placement; level statistics computation.
- **Out of scope:** Sub-daily (hourly/15-minute) timeseries; spatial correlation across gauges; rainfall–runoff modelling; real-time data assimilation; tidal harmonic analysis beyond the spring-neap envelope.

- **Upstream dependencies:** Gauge portfolio (`gauge.json`) providing flood thresholds, historical high levels, and geographic coordinates; CDM schema defining the `TidalInfluence` field.
- **Downstream consumers:** Stress model pipeline (`stressm`); GEV hazard curve builder (MKM-HAZARD-001); gauge timeseries visualisation; Trading Desk historical tab.

## 2.3 Regulatory Context

This model is classified as a *Tier 2* component under the internal model governance framework. It does not directly produce a regulated output but is a material input to the hazard curve and stress testing models, both of which are Tier 1. Accordingly, it is subject to annual independent validation and parameter review.

# 3 Mathematical Framework

## 3.1 Overview

The synthetic water level at day  $t$  is constructed as the sum of five components:

$$W_t = \mu_0 + S_t + T_t + \varepsilon_t + F_t \quad (1)$$

where  $\mu_0$  is the base level,  $S_t$  is the seasonal component,  $T_t$  is the tidal component,  $\varepsilon_t$  is the AR(1) autocorrelated noise, and  $F_t$  is the flood injection term. A floor of  $W_{\min} = 0.1$  m is applied to prevent physically meaningless negative levels.

## 3.2 Tidal Classification

Each gauge is classified into one of three tidal regimes based on its longitude  $\lambda$  relative to Teddington Lock, the tidal limit of the Thames:

$$\text{TidalInfluence} = \begin{cases} \text{Tidal} & \text{if } \lambda > -0.25 \\ \text{Partially tidal} & \text{if } -0.35 < \lambda \leq -0.25 \\ \text{Non-tidal} & \text{if } \lambda \leq -0.35 \end{cases} \quad (2)$$

The longitude thresholds are calibrated to the Thames geography. Teddington Lock sits at approximately  $\lambda \approx -0.32$ ; the classification bands provide a transition zone to account for partially dampened tidal influence in the reach immediately upstream.

The tidal classification is stored in the CDM field `SensorDetails.GaugeInformation.TidalInfluence` and determines the parameter set used for all subsequent model components.

## 3.3 Base Level

The base level  $\mu_0$  is drawn uniformly from a fraction of the FloodAlert threshold  $h_{\text{alert}}$ :

$$\mu_0 = h_{\text{alert}} \cdot U(0.3, 0.5) \quad (3)$$

where  $U(a, b)$  denotes a uniform random variable on  $[a, b]$ . This ensures the typical water level sits well below the alert threshold, consistent with observed gauge behaviour.

### 3.4 Seasonal Component

The seasonal variation captures the winter-dominant rainfall pattern in the Thames catchment. The seasonal component at day  $t$  is:

$$S_t = A_s \cos\left(\frac{2\pi(d_t - 15)}{365}\right) \quad (4)$$

where  $d_t$  is the day-of-year (1–365) at time  $t$ , and  $A_s$  is the seasonal amplitude. The phase offset of 15 days positions the peak in mid-January, reflecting the typical Thames winter high-water season. The seasonal amplitude varies by tidal type (Table 1).

### 3.5 Tidal Component

For tidally-influenced gauges, a spring-neap modulation is applied with a period of approximately 14.8 days (the synodic half-month):

$$T_t = A_\tau \cdot M_t \cdot U_t(-1, 1) \quad (5)$$

where  $A_\tau$  is the tidal amplitude,  $U_t(-1, 1)$  is an independent uniform draw representing the semi-diurnal tidal phase (aggregated to daily resolution), and  $M_t$  is the spring-neap modulation envelope:

$$M_t = 0.6 + 0.4 \cos\left(\frac{2\pi(d_t \bmod 14.8)}{14.8}\right) \quad (6)$$

The modulation oscillates between 0.2 (neap tides, minimum amplitude) and 1.0 (spring tides, maximum amplitude). For non-tidal gauges,  $A_\tau = 0$  and  $T_t = 0$  identically.

### 3.6 AR(1) Autocorrelated Noise

Daily water level fluctuations exhibit persistence: today's deviation from the mean is correlated with yesterday's. This is modelled as a first-order autoregressive process:

$$\varepsilon_t = \phi \varepsilon_{t-1} + \eta_t \quad (7)$$

where  $\phi \in (0, 1)$  is the autocorrelation coefficient and  $\eta_t$  is a Gaussian innovation:

$$\eta_t \sim \mathcal{N}(0, \sigma_w^2(1 - \phi^2)) \quad (8)$$

The innovation variance is scaled by  $(1 - \phi^2)$  so that the unconditional (marginal) standard deviation of  $\varepsilon_t$  equals  $\sigma_w$ , the daily noise standard deviation. This ensures that the total variability of the generated series is controlled by  $\sigma_w$  regardless of the autocorrelation strength.

The autocorrelation coefficient  $\phi$  varies by tidal type:

- **Non-tidal** ( $\phi = 0.85$ ): High persistence. Upstream river levels change slowly, driven by sustained rainfall and baseflow.
- **Partially tidal** ( $\phi = 0.75$ ): Moderate persistence. Tidal signal introduces some daily reset but rainfall effects still dominate.

- **Tidal** ( $\phi = 0.6$ ): Lower persistence. The semi-diurnal tidal cycle resets the level twice daily, reducing day-to-day correlation.

### 3.7 Flood Event Injection

Flood events are injected as Poisson arrivals at two severity levels:

#### 3.7.1 Severe Floods

Severe flood events occur with daily probability:

$$p_{\text{severe}} = \frac{\max(n_3, 1)}{N} \quad (9)$$

where  $n_3$  is the historical count of Level 3 exceedances (`FrequencyExceedLevel3` from the gauge portfolio) and  $N = 365 \times Y$  is the total number of days in the simulation ( $Y = 50$  years by default). On a severe flood day, the water level is drawn uniformly:

$$W_t \sim U(h_{\text{severe}}, 0.95 h_{\text{max}}) \quad (10)$$

where  $h_{\text{severe}}$  is the `SevereFloodWarning` threshold and  $h_{\text{max}}$  is the historical high level.

#### 3.7.2 Minor Floods

Minor flood events occur at five times the severe rate:

$$p_{\text{minor}} = 5 p_{\text{severe}} \quad (11)$$

and produce water levels in the alert-to-warning range:

$$W_t \sim U(h_{\text{alert}}, h_{\text{warning}}) \quad (12)$$

Both flood types reset the AR(1) state:  $\varepsilon_t \leftarrow W_t - \mu_0$ , allowing the autocorrelation process to decay naturally from the elevated flood level.

### 3.8 Historical High Date Injection

If the gauge record includes a `HistoricalHighDate` that falls within the simulation window, the model places the exact historical high level at that date:

$$W_{t^*} = h_{\text{max}} \quad \text{where } t^* = \text{HistoricalHighDate} \quad (13)$$

The AR(1) state is reset to  $\varepsilon_{t^*} = h_{\text{max}} - \mu_0$  so that subsequent days decay naturally from the peak. If the historical high date falls outside the simulation window, it is not injected.



### 3.9 Output Floor

All generated levels are floored at  $W_{\min} = 0.1$  m:

$$W_t \leftarrow \max(0.1, W_t) \quad (14)$$

This prevents negative or near-zero levels that would be physically unrealistic for a permanently gauged river reach.

## 4 Input Parameters and Calibration

### 4.1 Parameter Table by Tidal Type

Table 1 lists all tidal-regime-dependent parameters. All amplitudes and standard deviations are expressed as fractions of the base level  $\mu_0$ .

Table 1: Model parameters by tidal classification.

| Parameter                      | Non-tidal | Partially tidal | Tidal |
|--------------------------------|-----------|-----------------|-------|
| Daily noise $\sigma_w/\mu_0$   | 0.08      | 0.12            | 0.20  |
| Seasonal amplitude $A_s/\mu_0$ | 0.30      | 0.22            | 0.20  |
| Tidal amplitude $A_\tau/\mu_0$ | 0.00      | 0.06            | 0.15  |
| AR(1) coefficient $\phi$       | 0.85      | 0.75            | 0.60  |

### 4.2 Gauge-Specific Parameters

Table 2 lists the parameters extracted from each gauge’s CDM record.

Table 2: Per-gauge input parameters from the gauge portfolio.

| Parameter            | CDM Field             | Default                        | Description                       |
|----------------------|-----------------------|--------------------------------|-----------------------------------|
| $h_{\text{alert}}$   | FloodAlert            | 3.0 m                          | FloodAlert threshold              |
| $h_{\text{warning}}$ | FloodWarning          | 4.0 m                          | FloodWarning threshold            |
| $h_{\text{severe}}$  | SevereFloodWarning    | 5.0 m                          | SevereFloodWarning threshold      |
| $h_{\text{max}}$     | HistoricalHighLevel   | $1.1 \times h_{\text{severe}}$ | Historical high water level       |
| $t^*$                | HistoricalHighDate    | —                              | Date of historical high           |
| $n_3$                | FrequencyExceedLevel3 | 0                              | Level 3 exceedance count (5-year) |
| TidalInfluence       | TidalInfluence        | Non-tidal                      | Tidal regime classification       |

### 4.3 Global Parameters

Table 3: Global simulation parameters.

| Parameter                | Default   | Description                          |
|--------------------------|-----------|--------------------------------------|
| $Y$ (simulation years)   | 50        | Length of generated timeseries       |
| $W_{\min}$ (level floor) | 0.1 m     | Minimum water level                  |
| Phase offset             | 15 days   | Seasonal cosine peak (mid-January)   |
| Spring-neap period       | 14.8 days | Lunar half-synodic month             |
| Minor/severe ratio       | 5:1       | Ratio of minor to severe flood rates |

### 4.4 Tidal Classification Thresholds

The tidal classification (equation 2) uses the following longitude boundaries for the Thames catchment:

Table 4: Tidal classification longitude thresholds (Thames).

| Classification  | Longitude Range              | Physical Interpretation         |
|-----------------|------------------------------|---------------------------------|
| Tidal           | $\lambda > -0.25$            | Downstream of Teddington Lock   |
| Partially tidal | $-0.35 < \lambda \leq -0.25$ | Transition zone near Teddington |
| Non-tidal       | $\lambda \leq -0.35$         | Upstream of Teddington Lock     |

### 4.5 Calibration Approach

The tidal-regime parameters (Table 1) are calibrated by expert judgement informed by the following physical reasoning:

1. **Non-tidal gauges** are upstream and dominated by rainfall–runoff dynamics. Daily variability is low ( $\sigma_w/\mu_0 = 0.08$ ) because baseflow provides a stable level. Seasonal variation is large ( $A_s/\mu_0 = 0.30$ ) because winter rainfall is the primary driver. Autocorrelation is high ( $\phi = 0.85$ ) because river levels respond slowly to catchment inputs.
2. **Tidal gauges** are downstream and dominated by the semi-diurnal tide. Daily variability is high ( $\sigma_w/\mu_0 = 0.20$ ) due to the tidal range. Seasonal variation is smaller ( $A_s/\mu_0 = 0.20$ ) relative to the tidal signal. Autocorrelation is moderate ( $\phi = 0.6$ ) because the tide resets levels twice daily.
3. **Partially tidal gauges** interpolate between these extremes.

The flood injection probabilities are calibrated per-gauge from the historical Level 3 exceedance count in the gauge portfolio, ensuring that each gauge’s synthetic history reproduces the observed frequency of severe events.

## 5 Implementation Details

### 5.1 Software Architecture

The model is implemented across two layers:

- `src/models/statistics/synthetic.py` — the core `generate_synthetic_timeseries()` function containing all mathematical logic. Depends only on the Python standard library (`math`, `random`, `datetime`).

- `src/port/src/gauge/gaugehd/` — the generator package providing the `GaugeHistoricalDaily` facade class, portfolio loading, statistics computation, NRFA data parsing, and batch runner.

## 5.2 Module Structure

The `gaugehd` package comprises six modules:

- `__init__.py` — package facade, exports `GaugeHistoricalDaily` class.
- `generator.py` — `GaugeHistoricalDaily` class delegating to sub-modules.
- `synthetic.py` — `generate_from_gauge_portfolio()`: orchestrates timeseries generation, statistics computation, and JSON output.
- `statistics.py` — re-exports from `models.statistics.timeseries`.
- `nrfa.py` — NRFA GDF CSV parser for real gauge data ingestion.
- `runner.py` — batch generation (`generate_all_gauge_histories()`) and CLI entry point.
- `loader.py` — gauge portfolio loading from `gauge.json`.

## 5.3 Dependency-Free Design

The core generation function uses only the Python standard library:

- `math.cos`, `math.pi` for seasonal and tidal oscillations.
- `random.gauss` for AR(1) innovations.
- `random.uniform` for base level, tidal phase, and flood level draws.
- `random.random` for Poisson flood event arrivals.
- `datetime` for date arithmetic and formatting.

This eliminates deployment dependencies on NumPy or SciPy, consistent with the platform’s design philosophy.

## 5.4 Reproducibility

Each gauge’s timeseries is generated with a deterministic seed derived from its `GaugeID`:

$$\text{seed} = \text{hash}(\text{gauge\_id}) \bmod 2^{32} \quad (15)$$

This ensures that regeneration produces identical output for the same gauge portfolio, supporting audit reproducibility.

## 5.5 Output Format

Each gauge produces a JSON file `gauge-{gauge_id}.hd.json` in the `gaugehd/` directory with the following structure:

- `schema_version`: “1.0”
- `data_type`: “GaugeHistoricalDaily”
- `gauge_metadata`: gauge identity, coordinates, flood stages
- `statistics`: computed level statistics (mean, max, percentiles, monthly means, annual maxima, flood exceedance counts)
- `daily_observations`: array of `{date, level_meters}` records

## 5.6 Monthly Means Output

The statistics module computes monthly mean water levels from the generated daily observations. These monthly means are consumed by the stress model pipeline (`stressm/gauge_parser.py`) as the baseline water level for each month, against which stress scenarios are constructed. The monthly means inherit the seasonal pattern from equation (4), with January means approximately  $A_s$  above the annual mean and July means approximately  $A_s$  below.

## 6 Sensitivity Analysis

The model has four primary sensitivity dimensions: the AR(1) coefficient  $\phi$ , the seasonal amplitude  $A_s$ , the tidal amplitude  $A_\tau$ , and the daily noise standard deviation  $\sigma_w$ .

### 6.1 Sensitivity to AR(1) Coefficient ( $\phi$ )

The autocorrelation coefficient is the most influential parameter for the variance structure of the generated series. The lag- $k$  autocorrelation of the AR(1) component is  $\rho_k = \phi^k$ , so higher  $\phi$  produces longer-memory behaviour:

- At  $\phi = 0.85$  (non-tidal), the autocorrelation at lag 7 (one week) is  $0.85^7 \approx 0.32$ , meaning week-to-week persistence is significant.
- At  $\phi = 0.6$  (tidal), the lag-7 autocorrelation is  $0.6^7 \approx 0.028$ , meaning weekly persistence is negligible.

The innovation variance scaling  $(1 - \phi^2)$  ensures that the marginal variance is constant across tidal types for the same  $\sigma_w$ . However, the *effective* number of independent observations in the series decreases with higher  $\phi$ , amplifying sampling uncertainty in tail statistics.

### 6.2 Sensitivity to Seasonal Amplitude ( $A_s$ )

The seasonal amplitude directly controls the range of monthly mean levels. For non-tidal gauges ( $A_s/\mu_0 = 0.30$ ), the winter-to-summer swing is approximately  $0.6\mu_0$ . For tidal gauges ( $A_s/\mu_0 = 0.20$ ), the swing is  $0.4\mu_0$ . Since monthly means feed the stress pipeline, an over-estimated  $A_s$  will produce artificially low summer baselines that may distort stress scenario magnitudes.

### 6.3 Sensitivity to Tidal Amplitude ( $A_\tau$ )

The tidal amplitude affects only tidally-influenced gauges. At  $A_\tau/\mu_0 = 0.15$ , the spring-tide daily range is approximately  $0.3\mu_0$  (peak-to-trough), contributing substantially to the total variance. Since the tidal component uses a uniform random phase (not a deterministic tidal harmonic), it introduces additional variance but no systematic bias.

### 6.4 Sensitivity to Flood Injection Rate

The severe flood probability  $p_{\text{severe}}$  is calibrated from the gauge's historical Level 3 exceedance count. Gauges with  $n_3 = 0$  default to  $n_3 = 1$  (minimum one severe event in 50 years). The minor flood rate is always five times the severe rate. These rates are the primary control on the upper tail of the generated water level distribution.

## 6.5 Recommended Sensitivity Tests

1. Vary  $\phi$  from 0.4 to 0.95 in steps of 0.05 for a representative non-tidal gauge. Report the lag-1 to lag-30 autocorrelation function and the empirical flood exceedance frequency.
2. Vary  $A_s/\mu_0$  from 0.10 to 0.40 in steps of 0.05. Report monthly mean levels and the January/July ratio.
3. For a tidal gauge, vary  $A_\tau/\mu_0$  from 0.05 to 0.25. Report total daily variance and the fraction of variance attributable to the tidal component.
4. Set  $n_3 = 0, 1, 3, 5$  for a single gauge and compare the 1-in-50-year and 1-in-100-year return levels from the generated series.

## 7 Validation and Backtesting

### 7.1 Statistical Diagnostics

The following diagnostics are performed on each generated timeseries:

1. **Autocorrelation function (ACF).** The empirical ACF of the generated series should exhibit exponential decay consistent with the specified  $\phi$ . For a non-tidal gauge ( $\phi = 0.85$ ), the lag-1 autocorrelation should be approximately 0.85.
2. **Seasonal decomposition.** Monthly mean levels should exhibit a clear winter peak (January) and summer trough (July), with amplitude proportional to  $A_s$ .
3. **Flood frequency.** The number of days exceeding the FloodAlert threshold over the 50-year simulation should be consistent with the specified flood injection rates: approximately  $5 \times \max(n_3, 1)$  minor floods and  $\max(n_3, 1)$  severe floods.
4. **Historical high placement.** If `HistoricalHighDate` is within the simulation window, the generated level on that date must equal `HistoricalHighLevel` exactly.

### 7.2 Cross-Gauge Consistency

Across the 40-gauge Thames portfolio:

- Tidal gauges (downstream,  $\lambda > -0.25$ ) should exhibit higher daily variance than non-tidal gauges (upstream,  $\lambda \leq -0.35$ ).
- The seasonal amplitude ratio  $A_s/\sigma_{\text{total}}$  should decrease from non-tidal to tidal gauges, reflecting the increasing dominance of tidal variability.
- All gauges should produce mean levels below their FloodAlert threshold.

### 7.3 Reproducibility Verification

Given the deterministic seeding (equation 15), regeneration with the same gauge portfolio must produce byte-identical JSON output. This is verified by the automated test suite (`tests/port/gaugehd/`).

### 7.4 Backtesting Protocol

When NRFA real gauge data is available for a station, the synthetic timeseries should be compared to the observed record:

1. Compare empirical monthly mean profiles (synthetic vs. observed).

2. Compare the empirical exceedance frequency at the FloodAlert threshold.
3. Compute the Kolmogorov–Smirnov statistic between the synthetic and observed daily level distributions.

## 8 Model Limitations and Known Weaknesses

### 8.1 Simplified Tidal Model

The tidal component uses a single spring-neap envelope with uniform random phase, rather than a proper harmonic decomposition of the tidal constituents (M2, S2, N2, K1, O1, etc.). This captures the fortnightly amplitude modulation but not the precise timing or asymmetry of real tidal cycles. For downstream applications that depend on the timing of high water relative to storm surge, a full tidal harmonic model would be required.

### 8.2 No Spatial Correlation

Each gauge’s timeseries is generated independently. In reality, gauges within the same river reach share common hydrological drivers (rainfall, catchment moisture), producing strong spatial correlation. The independence assumption means that portfolio-level statistics computed from the generated series will under-state the co-occurrence of elevated water levels across multiple gauges.

### 8.3 Stationarity

All parameters are treated as time-invariant over the 50-year simulation. The model does not account for climate-change-driven trends in mean water levels, seasonal amplitude, or flood frequency. Users requiring non-stationary analysis should treat the generated series as representative of current-climate conditions.

### 8.4 AR(1) Limitations

The first-order autoregressive model captures day-to-day persistence but not longer-range dependence structures such as:

- Multi-day flood events (where levels remain elevated for several consecutive days due to sustained rainfall).
- Long-memory processes (Hurst exponent  $> 0.5$ ) that have been documented in some hydrological timeseries.

The AR(1) process produces single-day flood spikes; realistic multi-day flood events would require a higher-order AR model or a dedicated event model.

### 8.5 Uniform Flood Level Distribution

Flood event levels are drawn from uniform distributions (equations 10 and 12), which assigns equal probability to all levels within the flood range. In reality, lower flood levels are more common than extreme floods, and a GEV or log-normal distribution would be more appropriate. The uniform assumption over-states the frequency of the most extreme flood levels within each severity band.

## 8.6 No Baseflow Separation

The model does not separate baseflow (groundwater-driven) from quickflow (rainfall-driven) components. For non-tidal gauges where baseflow dominates the dry-season level, this is acceptable. For catchments with significant groundwater variability, the constant base level assumption may produce overly smooth summer levels.

## 8.7 Fixed Simulation Length

The default simulation length is 50 years. For return periods longer than 50 years (e.g. 1-in-100-year events), the generated series provides limited empirical information and downstream GEV fitting must extrapolate beyond the data range.

# 9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

## 9.1 Q1 — Ronseal Test

*“Does the model do what it says on the tin?”*

Yes. The model generates synthetic daily water level timeseries for flood gauges, producing 50-year histories with seasonal, tidal, and stochastic components calibrated to each gauge’s flood thresholds and tidal classification. The generated data is consumed by downstream models (hazard curves, stress testing) as documented. Monthly means, level statistics, and flood exceedance counts are computed as specified.

## 9.2 Q2 — Accuracy and Stability

*“Does the model provide timely, accurate and stable results?”*

The model produces deterministic output for a given gauge portfolio (via hash-based seeding). Generated levels respect the specified flood thresholds (base level is 30–50% of FloodAlert). The AR(1) innovation variance scaling ensures marginal variance is independent of the autocorrelation coefficient. Historical high levels are placed at their exact recorded dates. Results are stable across regeneration cycles.

## 9.3 Q3 — Resource Usage

*“Does the model consume excessive computational resources?”*

No. The model uses only the Python standard library with no external dependencies. Generating 50 years of daily data (18,250 records) for a single gauge completes in under 50 ms. The full 40-gauge Thames portfolio is generated in under 2 seconds. Memory usage is proportional to the number of daily observations (approximately 2 MB per gauge JSON file).

## 9.4 Q4 — Override Controls

*“Are there controls over manual overrides?”*

Model parameters are determined by the tidal classification (derived from longitude) and the gauge portfolio CDM fields. There is no mechanism for manual real-time override of generated water levels. The simulation length (50 years) and tidal classification thresholds are hard-coded constants. Changes require code modification subject to version control.

## 9.5 Q5 — Output Consumption

*“How is the output consumed for risk decision-making?”*

The generated timeseries serves three downstream purposes: (i) monthly means feed the stress model pipeline (`stressm`) for baseline water levels in stress scenarios; (ii) the full daily series provides the empirical distribution for GEV hazard curve fitting; (iii) level statistics (percentiles, annual maxima, flood counts) are displayed in the Trading Desk historical tab and gauge visualisation panels.

## 9.6 Q6 — Model Chaining

*“Is the model’s output used as input to another model?”*

Yes. The model is an upstream component in the model chain. Its primary downstream consumers are: (a) the stress model pipeline (MKM-SM-001) which uses monthly means as baseline levels; (b) the GEV hazard curve builder (MKM-HAZARD-001) which fits exceedance distributions to the generated daily levels; and (c) the gauge timeseries visualisation. The model’s only upstream dependency is the gauge portfolio (`gauge.json`).

## 9.7 Q7 — Weaknesses

*“Does the model have weaknesses and limitations of use?”*

Yes; seven limitations are documented in Section 8. The principal weaknesses are: (i) simplified tidal model (spring-neap envelope only, no harmonic decomposition); (ii) no spatial correlation across gauges; (iii) stationarity assumption (no climate trends); (iv) AR(1) cannot produce multi-day flood events; (v) uniform flood level distribution over-states extreme frequencies within each severity band.

## 9.8 Q8 — Monitoring

*“What type of monitoring is required?”*

Annual monitoring should include: (a) comparison of synthetic monthly mean profiles against NRFA observed data where available; (b) verification that flood exceedance counts are consistent with the gauge portfolio’s `FrequencyExceedLevel3` field; (c) ACF analysis confirming that the empirical lag-1 autocorrelation matches the specified  $\phi$ ; (d) cross-gauge consistency checks (tidal variance > non-tidal variance); (e) reproducibility verification via hash-based seeding.



## 9.9 Q9 — Overall Risk

*“Is the overall model risk acceptable?”*

Acceptable. As a Tier 2 model, the GaugeHD synthetic generator does not directly produce a regulated output but provides the historical baseline for Tier 1 downstream models. The key risk—tidal mis-classification distorting the variance structure—is mitigated by the deterministic longitude-based classification (equation 2). The independence assumption across gauges is acceptable because downstream portfolio-level risk aggregation uses separate spatial correlation models. The stationarity limitation is acceptable provided the model is recalibrated when climate projections materially change baseline water levels.

## 10 Change History

This section records all material changes to the model methodology, parameters, or implementation.

- **v1.0 (18-Mar-2026):** Initial model documentation. Covers the five-component additive water level model (base, seasonal, tidal, AR(1), flood injection), tidal classification by longitude, parameter tables by tidal type, and governance validation responses.

## 11 Document History

| Date        | Version | Description           | Author   |
|-------------|---------|-----------------------|----------|
| 18-Mar-2026 | 1.0     | Initial documentation | D. Kelly |